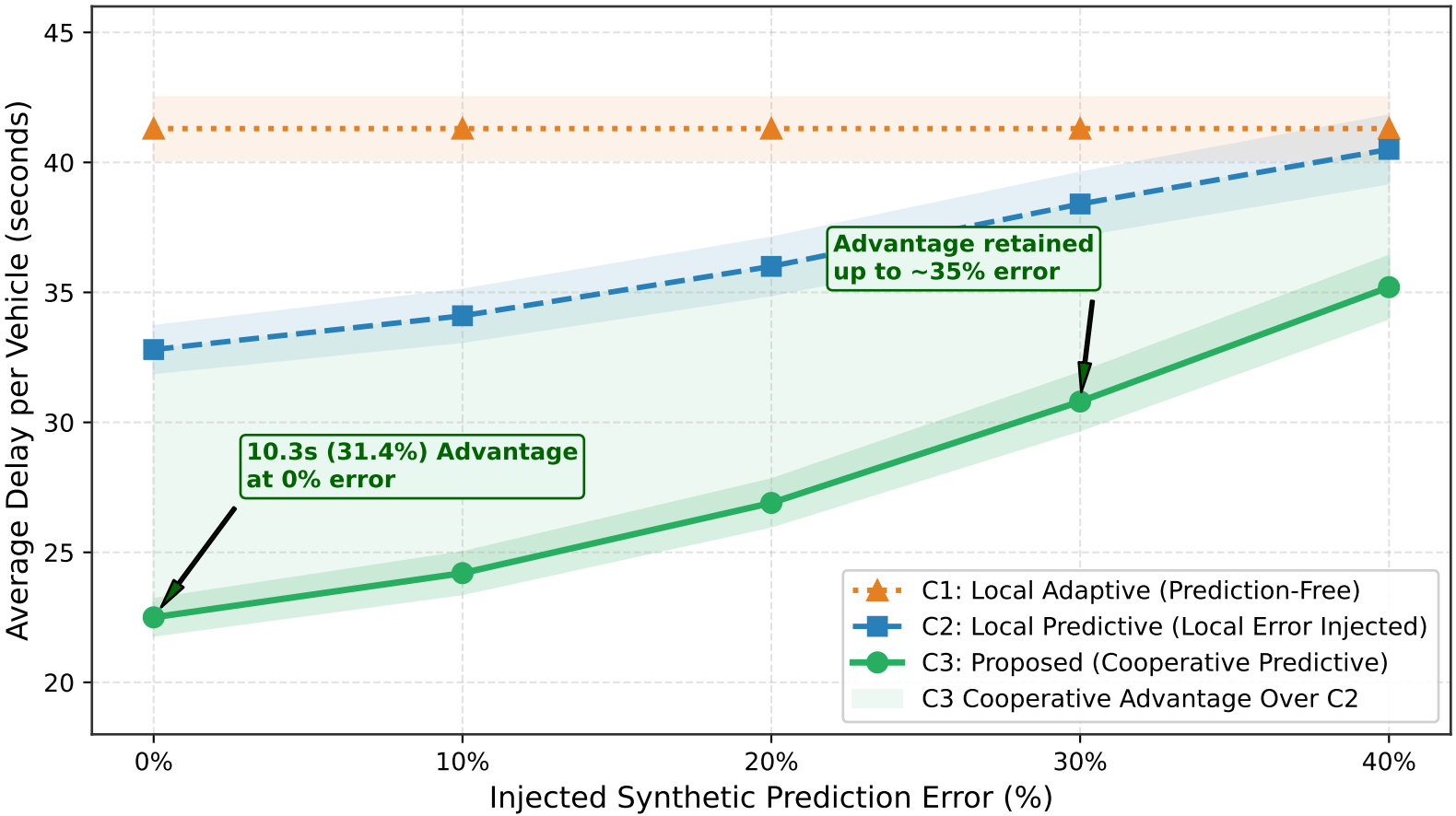
Prediction Error Sensitivity Analysis (RQ3)
Average Delay per Vehicle (seconds)
45
40
35
30
25
20
0%
10%
20%
30%
40%
Injected Synthetic Prediction Error (%)
Advantage retained
up to ~35% error
10.3s (31.4%) Advantage
at 0% error
C1: Local Adaptive (Prediction-Free)
C2: Local Predictive (Local Error Injected)
C3: Proposed (Cooperative Predictive)
C3 Cooperative Advantage Over C2